



## Understand bias in data

I may be biased, but I think learning about the good, and the bad traits of data, is pretty fascinating. Next up, we'll discover that there's lots of different types of data bias, besides sampling bias, which we covered earlier.

As a quick refresher, **sampling bias**, is when a sample isn't representative of the **population as a whole**. For example, if you're doing research on **commuters**, and only survey people walking by in the sidewalk, you'll miss out on input from people who ride bicycles, drive, or take the subway. You need all sides of the story to avoid sampling bias. **In this video**, **we'll explore three more types of data bias**, **observer bias**, **interpretation bias**, and **confirmation bias**, **and we'll learn how to avoid them**. Let's start with **observer bias**, **which is sometimes referred to as experimenter bias or research bias**. **Basically, it's the tendency for different people to observe things differently**. You might remember earlier, we learned that scientists use observations a lot in their work, like when they're looking at bacteria under a microscope to gather data. While two scientists looking into the same microscope might see different things, that's observer bias. Another time observer bias might happen is during manual blood pressure readings. Because the pressure meter is so sensitive, health care workers often get pretty different results. Usually, they'll just round up to the nearest whole number to

compensate for the margin of error. But if doctors consistently round up, or down the blood pressure readings on their patients, health conditions may be missed, and any studies involving their patients wouldn't have **precise**, and accurate data. Another common type of data bias is **interpretation bias**. **The tendency to always interpret ambiguous situations in a positive, or negative way.** Here's an example. Let's say you're having lunch with a colleague, when you get a voicemail from your boss, asking you to call her back. You put the phone down in a **huff**, certain that she's angry, and you're on the hot seat for something. But when you play the message for your friend, he doesn't hear anger at all, he actually thinks she sounds calm and straightforward. **Interpretation bias, can lead to two people seeing or hearing the exact same thing, and interpreting it in a variety of different ways, because they have different backgrounds, and experiences.** Your history with your boss made you interpret the call one way, while your friend interpreted it in another way, because they're strangers. Add these interpretations to a data analysis, and you can get bias results. The last type of bias we'll cover, reminds me of the saying, people see what they want to see. That pretty much sums up confirmation bias in a **nutshell**. **Confirmation bias, is the tendency to search for, or interpret information in a way that confirms preexisting beliefs. Someone might be so eager to confirm a gut feeling, that they only notice things that support it, ignoring all other signals.** This happens all the time in everyday life. We might get our news from a certain website because the writers share our beliefs, or we socialize with people because we know that they hold similar views. After all, conflicting viewpoints might cause us to question our worldview, which can lead us to changing our whole belief system, and let's face, it, change is tough. But you know what's even tougher? Doing

good work when you have bad data, so it's important to keep bias out of it. The four types of data bias we covered, **sampling bias**, **observer bias**, **interpretation bias**, and **confirmation bias**, are all unique, but they do have **one thing in common**. **They each affect the way we collect, and make sense of the data**. Unfortunately, they're also just a small sample, **pun** intended, of the types of bias you may encounter in your career as a data analyst. But the good news is, once you know a few, you'll find yourself constantly on guard for bias in any form. It's also important to remember, that no matter what kind of data you use, all of it needs to be inspected for accuracy, and trustworthiness. We'll talk more about that soon when we start exploring bad data. Bye for now.